Team 3

Does Anonymity Still Exist Online?

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Executive Summary

In an age where information technology is abundant and used in our everyday lives, how much privacy does one truly have? With the onset of the SARS-CoV-2 pandemic, it has become ever more prudent to think about the consequences of privacy in the online world as people have transitioned their lifestyles to be more centered around the web. Systematic surveillance of social media has become an increasingly recent and relevant concern as there are moral dilemmas on whether or not taking away the privacy of the general public is worth the benefit of public safety and security. Internet users have found ways to "protect" oneself by partaking in online anonymity, where their behavior and actions lead to lesser repercussions if used responsibly. Anonymous users have the ability to misuse this power and misbehave to an extreme extent, leading to consequences with the law.

However, these "anonymous" users might not be as hidden as they think. Companies and organizations with an interest in knowing the identities of individuals online have ramped up efforts in tracking with new methods such as browser fingerprinting and old technologies like cookies. These entities are using these methods in an attempt to track and associate users in an effort to personalize advertisements and keep tabs on browsing habits. To combat this, privacy tools such as VPNs and Proxies have emerged to service the growing concerns over privacy within the general public. While these tools do their best to protect identities online, they too can pose privacy issues as "no logging" promises have been found in many cases to be untrustworthy. Nothing is guaranteed but there are steps that can be taken to improve privacy, with the greatest step being to always take responsibility for your online actions.

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Introduction

With the increased use of technology over the years, we now live in a time where technology is widely implemented in many various aspects of our lives. One of the many uses of technology is that it's a great form of communication, allowing for ease of access to information. With so much accessible information and communication, communities and networks were created. These communities were mediated by technologies called "Social Media", which allowed people to share information amongst other people, whether it be personal details about your life or exchanging ideas with your friends. Even though these social media platforms are open to the general public, and you are knowingly putting yourself out there, social media users expect a degree of privacy when it comes to what they do or say on social media. Privacy concerns nowadays are increased as your actions on social media can lead to extremes such as legal repercussions and consequences. Alongside this, many techniques have been implemented by internet companies and world governments to track one's internet usage, whether in the name of protection or profit. So to what extent can you expect to have privacy on the internet, and how do you mitigate privacy violations?

Social Media

Social media has always been a complex topic or concept. To many, it is a way to communicate and to keep in touch with those you interact with. For some, it is an information hub where you have the ability to look up anyone you want to. There are users that want to exchange information with like-minded individuals, and there are others that stay anonymous and act on malicious intent. Social media mediates a platform where the general public willingly puts their information online, under the impression that it will be easier for others to find you, which can be beneficial or risky, a double-edged sword. Social media feels like a private space, but in actuality, it isn't. So you should operate under the impression that you have no privacy. Which also brings up the discussion of whether or not privacy exists in public areas.

Information is a very powerful tool, now more than ever, in an age where information is so accessible and abundant. The plethora of information over the internet and the use of technology means that there must be some way we can use that information to our advantage. With the goal of analyzing this information, two concepts, "Big Data" and "SOCMINT" were born. "Big Data" is a term used to describe data that can be analyzed and broken down to statistics, trends, facts, etc. Big data companies, such as Google, IBM, and other social media companies use this information to make profit in the form of marketing, advertising, etc. "SOCMINT" or "Social Media Intelligence" is information that is relevant to the government or military. This intelligence would potentially help the authorities predict certain actions or events that would take place, and overall help with our security. Both of these techniques bring up different privacy concerns. Is collecting information on what you type into a search engine for the sake of business profits a privacy breach? Are intelligence gathering techniques conducted by the authorities intruding on your privacy?

If you have ever paid for something online or used search engines to look something up, chances are, your information is a part of the giant data collections companies use to profit off of. If your information is used for scientific means, such as astrological or scientific reasons, then your personal information is probably not being used. However, in most cases, your personal information is being used and research is conducted by these Big Data companies as they analyze these data sets and create profiles of us (Smith, 2012). What they do with these profiles is completely up to their intentions, but a common occurrence that we see are targeted advertisements. These targeted advertisements cater to our interests based off of what we search, buy, and maybe even say. Efforts to place regulations on what information these companies can acquire are being increased as concerns entail the issue of government surveillance. It is very difficult to be aware of everything that affects us in terms of what data we provide and what data these companies are using just because of the sheer number of data being used by these companies. That is why we must combat the potential misuse of this data by being aware of this issue and attempt to regulate big data.

The government has always employed intelligence-gathering techniques and dedicated organizations for the sole purpose of acquiring and intercepting this information throughout the years. With the introduction and influx of technology, the authorities also adapted, which resulted in the creation of Social Media Intelligence (SOCMINT). Regardless of the ever-changing environment, is there privacy in public spaces? Authorities in charge of security and safety use social media as information hubs. Photos and posts across the platform can help identify individuals and timestamp events. Premeditated crimes can be stopped if evidence is found across social media (Rønn, 2019). SOCMINT is very beneficial towards the public safety and security of our society, however, government authorities acquiring this information through social media platforms would be the equivalent of surveillance of public spaces. There is a valid argument on consent when it comes to social media. Being on these social media platforms means that you consent to providing this information to the platform and that you consent to that information being public via your privacy settings.

For now, social media plays a huge role when it comes to information gathering and analyzing. Do not expect any of your online actions or words to be private, especially since there are companies and organizations that benefit from that data. Overall, this is something that we have to accept as there are not many legal repercussions or rights that are being violated. Maybe the circumstances surrounding big data and SOCMINT will change as society progresses. Just be aware that for now, these things exist and to act accordingly.

Anonymity

The ongoing debate about a person's right to anonymity and whether a person's real name and picture is what qualifies as social media identity has not reached a concrete conclusion. Anonymous postings have been around since 1988, in which they were primarily used for newsgroups when discussing sensitive and controversial topics. Those ways have evolved in today's society, as the Internet and social media usage has grown over the many years. There are now two types of anonymity. The definition of "anonymity" is that an author's name or identity is unknown and untraceable. The second term is "pseudonymity," in which the author uses

another name in place of their own. As mentioned, there are many arguments in regard to one's privacy rights. Some argue that anonymity is vital in protecting one's freedom of expression, while others argue that one should have enough conviction in their beliefs to share without hiding behind anonymity.

The argument for anonymity is indeed stemmed from the idea that it is in some way, protected by the First Amendment. This has been evident in the case of the New York Times Co. vs. Sullivan in 1954, in which the court stated that "an author's right to remain anonymous…is an aspect of the freedom of speech protected by the First Amendment" (Rigby, 1995). Our society today has become increasingly opinionated, in which certain topics are too sensitive to be talked about online. These topics include racism, abortion, minority issues, etc. People have a hard time expressing themselves because they do not want to be ridiculed for their own opinions and thoughts. There is an online debate website such as greatdebator.com, in which people can discuss controversial topics without feeling oppressed. Alongside these debate websites, doctors have also recommended anonymous support groups and forums where patients can share about their journey with others (Rigby, 1995). An example of this would be Alcoholics Anonymous, in which its members are encouraged to stay sober and help other alcoholics achieve sobriety (AA, 2021). A key point in argument for anonymity is that it has now become a part of society and almost unavoidable. It has become extremely accessible for one to create hundreds of fake accounts with fake names for different social media platforms. These are all forms of pseudonymity and will continue to be prevalent in our technological society.

However, it is important to note that anonymity does not mean you should hide your identity or information in an irresponsible manner. We must always be cautious of what is posted online. It is also important to consider the consequences of our actions as it could affect one's reputation. With the given freedom of expressing our speech and expressions, sometimes people take it to a certain extreme. In a second of posting something viscous or offensive online, it is never really gone forever (Dominic, 2021). A quick screenshot can later touch hundreds of people's phones and can further re-emerge again in the future. In Jennifer Bridges article, she talks about how there is a connection between online anonymity and civility. She goes into detail about online inhibition and how it has an effect on people's behavior online. According to John Suller, a Stanford psychology professor, the online inhibition effect can cause people to act out by oversharing personal information or become abusive/threatening (Bridges, 2019). There are several different elements that contribute to the disinhibition effect. The first is the dissociative anonymity, in which communication on the Internet being completely anonymous can make oneself feel protected. There is also invisibility, in which because everything is online, there are no concerns for one to worry about their tone of voice or body language. Last but not least there is solipsistic introjection, in which the other person that is communicating online becomes a "voice" in one's head and further lowers their own inhibition.

There are distinctive pros and cons to anonymity and the internet. It is a widely available tool for communication and allows people to share their thoughts and feelings freely without fear of being judged. However, with this privilege, anonymity can also be dangerous. If not used cautiously, there can be serious consequences to one's image and reputation when applying to jobs, schools, etc. This is why there have been verification systems in which online users can

verify their real identity. In this case, people are held accountable for bad behavior or mishaps in the online world. There are also other methods such as fingerprinting and tracking that can also be used to monitor online activity, which will next be further discussed.

Fingerprinting

A growing concern in the privacy space is the increasingly aggressive methods of tracking employed by companies and organizations that have interests in collecting as much data as they can from people threatening anonymity on the internet. Usually this data collection is relatively benign and for targeting things like ads and user suggestions. Yet, some are indeed a bit more nefarious and are used for building profiles to be able to uniquely identify particular users that are used by government entities for spying and observation of their citizens. There was once a time in the relatively near past that no tracking nor information was able to be collected, but a gradual increase in web browser capabilities have allowed for things like rudimentary cookies to be used for simple tracking. However, as technology has progressed, a new, more powerful method of tracking, browser fingerprinting, has entered the landscape, and its effectiveness has both been incredible as they are terrifying in their robustness and potential for abuse.

To begin with, we need to clarify what we mean when we are talking about fingerprinting. We are not talking about fingerprinting with your actual fingers (thankfully Apple's IOS and Google's Android operating systems do not allow for directly interfacing with fingerprint sensors). Rather, we are discussing the amalgamation of information that can be collected from your browser that could be used to identify you uniquely amongst any other user within a data set. Browsers are able to collect a lot of information such as the fonts and plugins you have installed, your time zone, your canvas, your JavaScript engine provided by your browser, and certain settings you have set like allowing cookies (Gómez-Boix et al., 2018). That information, like the creases on your fingers, are not very useful by themselves for identifying someone, but in whole, they likely form a signature that is unique to someone.

Some of the more common techniques for information collected to be used for fingerprinting involves the canvas API and your browser's JavaScript engine. HTML5 was released in 2014 as a powerful new web standard, a more flexible implementation of the HyperText Markup Language used to populate the content and dictate rendering of pages. With this new standard came the ability to manipulate the browser canvas using the canvas API provided by HTML5. This API is typically used to draw graphics and animations on areas of the browser screen, but companies and organizations have nefariously devised a method to use this feature to track users. This method entails drawing a small pixel on the screen and hashing that pixel using the user's computer. This hash has been approximated to have 10 bits of entropy which can be used to uniquely identify someone, and in conjunction with other information, it is a powerful filter for making unique fingerprints (Upathilake et al., 2015). In addition to this another method used relies on the differences between browsers and versions to identify users. JavaScript is a programming language that essentially powers the internet nowadays, and its engine is implemented in all major browsers with each implementation having a degree of nuance to them making them not all exactly the same. These nuances between browser's engines

and the different versions implemented in browsers leading to different capabilities makes them useful for fingerprinting. Many trackers could and do use methods to test a user's browser with tests targeted at finding out what specific browser is being used and what version of that browser you're on. This method is favored over using user-agent strings provided by the browser because it is far more robust and cannot be denied nor faked (Upathilake et al., 2015). These two common information gathering techniques are only but a few of the multitude of identifying information that's accessible to trackers attempting to build a fingerprint. The sheer amount of information that can be collected means being unique is far more difficult as any one factor could make you unique amongst everyone else's fingerprint.

It may seem perplexing to think of how to combat this invasion of privacy. How can you possibly stop all these different vectors that could be used to identify you? Unfortunately, there is no clear solution at the moment that can stop all these attempts to fingerprint you other than refusing to use the internet. However, that does not mean there aren't steps you can't take to be less unique among everyone else. Many of these methods for fingerprinting require JavaScript to run, so Ad Blockers such as uBlock Origin (note that Origin is open source, and thus preferred) that have filter lists that block tracking and fingerprinting scripts are effective in reducing attempts of fingerprinting you before they even start. There are also privacy minded-browsers such as Brave that have been developed to randomize information that is used to fingerprint you, so that you are always unique but never have the same fingerprint between sessions. Other browsers like Firefox attempt to mask your fingerprint information and make you blend in with the rest of the users to protect you. Using an ad blocker or a privacy minded browser, though won't protect you completely, is a good step for the average user to take in moving towards privacy minded browsing and defending against tracking and fingerprinting.

Privacy Tools

After learning about the extensive tracking practices and privacy violations of major companies, one major question rings in the mind of the privacy-minded individual: what can be done to preserve my anonymity online? Luckily, there are many solutions that exist to protect the average user. These can range from VPNs to better internet usage. While the privacy tools and protocols covered in this section have been around for over a decade, only recently have they been introduced in an easily accessible way to the public. This rapid advancement of anonymity technology is mostly due to the opinion of the public. In 2014, when asked if it is easy to be anonymous online, only 24% of adults agreed, with a resounding 74% of adults disagreeing (Madden). Perhaps the most concerning part of this study is that it was conducted in 2014, when the Big Data industry did not have the tools or computing capabilities of today. Seven years later, the online advertising industry is expected to become a $275 billion dollar industry in the next three years (Trefis), bringing along with it the privacy issues that present themselves in almost every facet of technology. One must first learn exactly how their privacy is being breached before they can learn how to combat it. This section will cover two different privacy tools that will aid in preserving one's privacy: HTTPS and VPNs.

HTTPS stands for Hypertext Transfer Protocol Secure. While the “secure” speaks for itself, the HTTP portion should be familiar to anyone who has used a web browser as the four characters that precede the URL in the address bar. When an application sends out an HTTP request, it is done so in a “plain-text” format. This means that should there be anything between the client and the server, known as the middleman, they will easily be able to read the contents of that request (CIO). In many cases, malicious actors or privacy breaching applications utilize the weakness of HTTP to be able to read exactly what a user is browsing or doing on a website. Anything from user data to passwords can be stolen in an HTTP transaction. To solve this, HTTPS has been adopted as a secure standard to ensure no entity between client and server can see what the client is doing. HTTPS utilizes the TLS security standard to encrypt requests, creating a secure environment for browsing the internet (Cloudflare). Luckily, the HTTPS standard has already been widely adopted on the internet, with the vast majority of servers for popular websites and apps using this standard. There will be many instances, however, when an individual has to use a website, whether it be for school or employment, that still runs on old software using the base HTTP standard. To overcome this, tools exist to force websites to run in HTTPS. The most popular is the “HTTPS Everywhere” extension for Chromium-based browsers. Tools like this will send a specific instruction to the server to run in HTTPS, avoiding any issues where a server might default to HTTP. These tools also include functionality to outright refuse connections if HTTPS is not used and is also built into some web browsers by default. While HTTPS aids user privacy by stopping privacy breaches by middlemen, it does not protect against companies that utilize location and IP address data to build a profile on a user. To overcome this, a VPN is the most effective form of privacy.

Many times, an individual will find themselves browsing the internet on a certain topic, only to go to a completely unrelated site and find advertisements about that topic. This is due to advertising companies utilizing the user’s IP address to build a profile on them. The more a user researches a certain topic or product, information is sent to these companies which utilize powerful algorithms to process the data and store their findings about what your IP address seems to like. An IP address itself can reveal plenty of information alone, including location and internet provider, but when combined with these profiles, companies and government agencies can see a clear path of interests, disinterests and the overall history of an individual on the internet. To stop this, one must utilize a Virtual Private Network, or VPN for short. A VPN encrypts all internet data and routes it through a secure server. The destination server only sees the IP address and information of the VPN’s server, so any attempts at building a profile on someone are thwarted. When combined with proper browser maintenance (clearing cookies at a regular rate), this will protect a user against advertising companies, monitored networks and governments. Using a VPN isn’t a new concept, and it is a tool that is already being widely adopted with 31% of internet users relying on a VPN. The VPN industry itself is expected to grow to an astounding $31.1 billion dollars (Arežina), meaning that more advancements in the technology and more servers to utilize are being created every day. Some caution must be used however, as a rapidly expanding industry can also bring in malicious actors. This can be seen in the Android VPN market, with 18% of VPN applications on the Google Play store not utilizing the proper encryption methods to ensure privacy (Ikram). While this seems to present an issue, it is easily dealt with by using proper research into VPN companies and reputations. By providing completely encrypted and anonymous web browsing that will protect against data hungry

companies and government regimes, **a VPN is not just the best tool to use on the internet: it is a must.**

Conclusion

Even with a VPN, corporations and organizations are always improving their tracking methods and data mining techniques to circumvent user attempts to remain anonymous. On the other hand, privacy tools and methods are also evolving and are being adopted at a rapid pace in the ongoing war of privacy on the web. Privacy has become an infinite cat-and-mouse game. While there will always be tools to protect a user, there will always be ways for the powers that be to acquire knowledge if it is needed enough. In short, true anonymity does not exist on the internet. However, the privacy tools and methods discussed above are effective enough to where the average internet user should not be paranoid about privacy violations in their day-to-day usage.

Author Biographies

Eunice Park is currently in her senior year studying Psychology at George Mason University. She has studied the human mind and why people think and act the way they do. With this knowledge, Eunice was able to dig deeper and look past the technology, and instead focused on the human population's role with social media and privacy. She is also an avid user of VPN services and different social media platforms, thus is familiar with their roles and how they continue to play a part in our society today.

Hrushikesh Ayalasomayajula is a junior Computer Science student at George Mason University. He has experience working in the Big Data department of multiple Fortune 50 companies, focusing on automated AWS file validation and error handling. In middle school, he used to create and sell bots that worked primarily on social media. He is experienced with web technologies, and currently focuses on web application development and deployment.

Kenny Peng is a senior Computer Science student at George Mason University. He is an experienced software engineer that has worked at multiple different companies doing DevOps and web development work. He has experience with numerous different frameworks and web technologies used in websites and understands the type of privacy concerns users have and the current capability of today's technology for tracking. Throughout his various roles, he has picked up a general understanding for why and how companies attempt to invade user's privacy.

Tee Huynh is a senior studying Software Engineering at George Mason University. He is experienced with multiple software engineering projects and is researching web development. Tee is also familiar with organizational and industrial psychology, which helps with the understanding of how people behave in the professional workplace or. He is also familiar with the use of social media, anonymity, and privacy tools such as vpns, by interacting with them in his everyday life.

PRIVACY STATISTICS YOU SHOULD KNOW:

86%

86% of internet users have taken steps online to remove or mask their digital footprints

FINGERPRINTING & PRIVACY TOOLS

77%

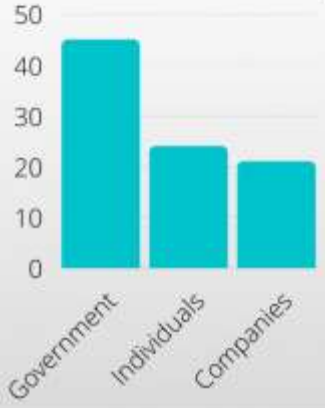
of all VPN users buy digital content every month

35.7%

of browser fingerprints from computers are unique

18.5%

of mobile device's browser fingerprints are unique



- 45 % of respondents indicated that they find the federal government responsible for protecting data privacy
- 24% of respondents find the individual user responsible for protecting data privacy
- 21% of respondents find that companies should be responsible for protecting data privacy

Cisco Consumer Privacy Survey 2019

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